WEEK 2 · LESSON 10 OF 24 · HANDOUT

Unsupervised Learning

Study time: **about 45 minutes** (reading, video and activity)

▶ **Watch:** https://www.youtube.com/watch?v=4b5d3muPQmA — open the lesson online to watch inside the course.

🎓 **Go deeper:** Harvard CS50 AI, Lecture 4: Learning (supervised, reinforcement, unsupervised) — https://www.youtube.com/watch?v=E4M_IQG0d9g

Your goal in this lesson: Explain what unsupervised learning is, describe clustering with k-means, and name two other unsupervised tasks used in business.

Learning without answers

In **unsupervised learning** the data has **no labels**. Nobody tells the computer the right answer. Instead, it looks for **structure**: groups, patterns and unusual points. It is like walking into a market for the first time and noticing, without being told, that the shops naturally group into vegetables, clothes and electronics.

Unsupervised learning is useful when labels are expensive, unknown, or when you want to *discover* something new.

Clustering

Clustering puts similar examples into groups (clusters). A popular method is **k-means**:

1. Choose *k*, the number of groups you want (say 3).
2. Place 3 starting centre points at random.
3. Assign every example to its nearest centre.
4. Move each centre to the middle (average) of its examples.
5. Repeat steps 3–4 until the groups stop changing.

Example: a telecom company clusters customers by calls, data use and mobile-money activity. It may discover:

Cluster	Pattern	Possible offer
1	Heavy data, few calls (students)	Night data bundles
2	Many calls, little data (traders)	Cheap call minutes
3	High mobile-money use (shop owners)	Merchant services

Nobody labelled these groups — the data revealed them. People then **name and interpret** the clusters; that part still needs human judgement.

Choosing k

There is no single correct *k*. Teams try several values and check whether the groups are **clearly separated** and **useful for a decision**. Three clusters you can act on are better than ten you cannot explain.

Other unsupervised tasks

- **Anomaly detection** — learn what "normal" looks like, then flag what is far from normal: a strange login, a machine about to break, an unusual payment.
- **Dimensionality reduction** — squeeze many features into a few while keeping most of the information (a common method is **PCA**, principal component analysis). It helps draw data on a chart and speeds up other models.
- **Association rules** — find items that often appear together: "people who buy bread often buy milk". Shops use this for shelf layout and offers.

Supervised vs. unsupervised

	Supervised	Unsupervised
Labels?	Yes	No
Goal	Predict a known answer	Discover structure
Typical tasks	Classification, regression	Clustering, anomaly detection, dimensionality reduction
How to judge	Compare with true labels	Usefulness and clarity, judged by people

There is also **semi-supervised learning** (a few labels plus lots of unlabelled data) and **self-supervised learning**, where the data creates its own labels — for example, hiding a word in a sentence and learning to guess it. Self-supervised learning is how large language models are pre-trained (Lesson 17).

Remember this: Unsupervised learning finds structure in unlabelled data — groups, unusual points and simpler summaries. People still decide what the patterns mean.

Name: _____ Date: _____

Activity

Task 1

Write down 12 foods you ate this month with two features each: price (low/high) and how healthy (low/high).

Task 2

Group them by hand into 3 clusters. Give each cluster a name.

Task 3

Change *k* to 2 and to 4. Which number gives the most useful groups, and why?

Self-check

Think about your answers here, then take the quiz online — it is marked instantly and explains every answer.

1. What is the main difference in unsupervised learning?
   - A. It needs no computer
   - B. The data has no labels
   - C. It only predicts prices
   - D. It always uses images
2. In k-means, what does k stand for?
   - A. The learning speed
   - B. The number of clusters you want
   - C. The number of features
   - D. The number of labels

3. After k-means assigns points to centres, what happens next?

A. The model is deployed
B. Each centre moves to the average of its points
C. Labels are added by hand
D. The data is deleted

4. Flagging an unusual mobile-money payment without labelled fraud examples is…

A. Linear regression
B. Translation
C. Anomaly detection
D. Recommendation

5. 'People who buy bread often buy milk' comes from…

A. Logistic regression
B. Association rules
C. Search
D. Reinforcement learning

6. Who decides what each discovered cluster means?

A. The customers vote
B. People interpret and name the clusters
C. The algorithm names them perfectly
D. Nobody — clusters have no use

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